THE COMPLETE BUSINESS KNOWLEDGE ENCYCLOPEDIA

الموسوعة الشاملة للمعرفة في عالم الأعمال

VOLUME 5

Commercial Operations — Sales, Supply Chain, Insurance and Risk

العمليات التجارية — المبيعات وسلاسل الإمداد والتأمين والمخاطر

Every Essential Business Concept, Term, Process, Document, Financial Instrument, and Commercial Practice You Need to Understand the Real Business World

IN THIS VOLUME

- Part VIII — Sales, Marketing and Commercial Strategy
- Part IX — Procurement, Logistics and Supply Chain
- Part XV — Insurance and Risk Management

A practical reference for finance and computer-science graduates entering the business world · First edition — October 2026

About this volume

This volume is part of a ten-volume reference collection. Chapter numbers run continuously across the collection, so a reference such as “§47.3” identifies the same section in every volume, in the master index (Volume 10), and in the learning roadmap.

Volume	Title
Vol. 1	Foundations and the Language of Business
Vol. 2	How Businesses Work — Accounting and Financial Statements
Vol. 3	Corporate Finance, Economics and Business Mathematics
Vol. 4	Banking, International Trade and Treasury
Vol. 5	**Commercial Operations — Sales, Supply Chain, Insurance and Risk**
Vol. 6	Investment, Capital Markets, Commodities and Market Intelligence
Vol. 7	Law, Entrepreneurship, Management and Technology
Vol. 8	Professional Practice — Communication, Deals and Advanced Thinking
Vol. 9	Document Library, Master Reference, Exercises and Learning Roadmap
Vol. 10	Master Table of Contents, Glossary and Unified Index

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PART VIII

Sales, Marketing and Commercial Strategy

المبيعات والتسويق والاستراتيجية التجارية

Revenue does not happen by itself. Someone must understand which customers to serve, what they value, how to reach them, how to convince them, at what price, on what terms, and how to keep them. This part covers the commercial engine of a business — marketing, sales, pricing, negotiation, account management and channels — with an emphasis on B2B practice, where most finance and technology graduates will work.

The chapters include practical tools: segmentation grids, qualification frameworks, discovery-meeting guides, sales scripts, emails, a full commercial proposal, negotiation preparation sheets and exercises.

LEARNING OBJECTIVES · **Learning objectives for Part VIII**

- Segment a market, choose targets and define a positioning and value proposition.
- Measure and manage acquisition, conversion, retention and customer economics.
- Run a B2B sales process from prospecting to signature, including discovery, proposals and closing.
- Price, discount and structure commercial proposals that protect margin.
- Prepare and conduct commercial negotiations.
- Manage key accounts, use CRM data, forecast sales and understand revenue operations.
- Choose and manage channels: direct sales, distributors, agents, resellers and partners.

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CHAPTER 62

Markets, Customers and Positioning

LEARNING OBJECTIVES · In this chapter

Segmentation, targeting, positioning, branding and value propositions — the strategic choices that precede any sales activity.

62.1 Segmentation

Segmentation divides a market into groups whose needs and buying behaviour differ enough to justify different offers, messages or channels. A good segment is **measurable**, **substantial**, **accessible**, **differentiable** and **actionable**.

Basis	B2C examples	B2B examples
Demographic / firmographic	Age, income, household	Industry, company size, ownership (government, private, multinational)
Geographic	Region, urban/rural	Country, cluster, free zone
Behavioural	Usage frequency, loyalty, channel	Purchase volume, buying process (tender vs direct), switching history
Needs-based	Convenience, status, price	Speed, reliability, technical support, credit terms
Value	Lifetime value	Profit potential, strategic value

WORKED EXAMPLE · Needs-based segmentation for a logistics company (fictional)

Segment	Need	Price sensitivity	Service model	Profit potential
Large retailers	Reliability, integration, scale	High (tender-driven)	Dedicated key-account teams, EDI/API	Medium (volume, thin margins)
Industrial importers	Customs expertise, project cargo handling	Medium	Technical sales, account managers	High
E-commerce SMEs	Simplicity, speed, digital booking	High	Online platform, self-service	Medium if automated
Pharma	Temperature control, compliance	Low	Specialist team, certifications	Very high

62.2 Targeting

Choose segments based on **attractiveness** (size, growth, margins, competition) and **ability to win** (fit with capabilities, reputation, cost position). Strategies: **undifferentiated** (one offer for all), **differentiated** (different offers per segment), **concentrated/niche** (focus on one or few segments). Small firms usually win with focus.

62.3 Positioning

Positioning is the place an offer occupies in the target customer's mind relative to competitors. A positioning statement:

> *For **[target segment]** who **[need]**, **[brand]** is the **[category]** that **[key benefit]**, because **[reason to believe]**, unlike **[main alternative]**.*

Example: “For industrial importers in Saudi Arabia who lose money on customs delays, Delta Clear is the customs and logistics partner that clears containers in 48 hours, because our licensed brokers and digital pre-clearance work with every port, unlike general forwarders who subcontract clearance.”

62.4 Branding

A brand is the set of associations and expectations attached to a name. In B2B, brand reduces perceived risk (“nobody gets fired for choosing a known supplier”), shortens sales cycles and supports price premiums. Brand assets include name, logo, reputation, certifications, references, thought leadership and the behaviour of every employee customers meet.

62.5 Value propositions

Use the customer’s economics. A value proposition is credible when it is **specific, quantified, relevant to the buyer’s role and provable**.

WORKED EXAMPLE · From features to value (fictional software for freight forwarders)

Feature	Benefit	Quantified value for a forwarder filing 1,500 declarations a month
Automatic data extraction from invoices	Less manual typing	1,500 × 12 minutes saved = 300 hours/month ≈ 2 FTEs ≈ $5,000/month
HS-code suggestions with audit trail	Fewer classification errors	Avoided penalties and delays, e.g., 3 fewer incidents × $2,000 = $6,000/month
Integration with customs single window	Faster submission	Containers released one day earlier → lower demurrage
Total value		**> $11,000 per month vs price $1,500 per month**

بالعربية — التجزئة والاستهداف والتموضع

تبدأ الاستراتيجية التسويقية بتقسيم السوق إلى شرائح ذات احتياجات مختلفة، ثم اختيار الشرائح التي تستطيع الشركة خدمتها بربحية وتميز، ثم تحديد المكانة التي تريد أن يحتلها المنتج في ذهن العميل مقارنة بالمنافسين. وعرض القيمة القوي يُعبَّر عنه بأرقام تخص اقتصاديات العميل نفسه: كم يوفر عليه، وكم يقلل من مخاطره، وكم يزيد من إيراداته.

PRACTICE · Practice questions

1. Segment the market for a B2B cleaning-services company in a Gulf city and choose one target segment with justification.
2. Write a positioning statement for a Lebanese olive-oil exporter targeting premium European retailers.
3. Convert two features of a cloud accounting system into quantified value for a 30-person trading company.

ANSWER · Answers to the practice questions

1. Segments: offices (multi-tenant towers), hospitals/clinics, hotels, schools, industrial sites, residential compounds. A plausible target: healthcare facilities — higher willingness to pay for

compliance and infection control, recurring contracts, fewer low-cost competitors, provided the company can obtain certifications and trained staff.

2. “For premium European grocers whose customers seek authentic single-origin products, [brand] is the extra-virgin olive oil from century-old Lebanese groves that offers traceable harvest-to-bottle quality, certified by independent labs, unlike blended oils of unclear origin.”
3. Bank-feed automation: reconciliation time cut from 3 days to half a day per month → ~20 hours saved × $25 = $500/month. Real-time multi-currency reporting: faster month-end close (5 days earlier) → management decisions on pricing and stock earlier; avoided FX errors. Quantify against subscription cost.

CHAPTER 63

Customer Acquisition, Retention and Commercial Metrics

LEARNING OBJECTIVES · In this chapter

How customers are acquired and retained, the funnel and its metrics, customer economics (CAC, LTV, payback), retention and churn analysis.

63.1 Acquisition channels

Channel	B2B examples	B2C examples	Typical economics
Outbound sales	Prospecting calls, emails, LinkedIn, events	Door-to-door, telesales	High cost per customer; suits high-value deals
Inbound marketing	Content, SEO, webinars, case studies	Search, social content	Lower cost at scale; slower to build
Paid acquisition	Search ads, LinkedIn ads	Search, social, display, influencers	Scalable; CAC can rise quickly
Referrals and partnerships	Introductions by banks, accountants, forwarders	Referral programmes	Low CAC, high trust
Tenders and marketplaces	Government and corporate tenders, procurement platforms	Online marketplaces	Price-competitive; fees
Trade shows	Industry exhibitions	Consumer fairs	Expensive; good for relationships in some industries
Distributors and resellers	Channel partners	Retail chains	Partner margin; reach

63.2 The funnel and its metrics

WORKED EXAMPLE · A B2B funnel (fictional SaaS for distributors, quarterly)

Stage	Volume	Conversion to next
Website visitors from target accounts	6,000	2.0%
Leads (demo requests, downloads)	120	50%
Sales-qualified leads (SQL)	60	40%
Opportunities (discovery done, need confirmed)	24	50%
Proposals sent	12	42%
Closed-won	5	—

To win 10 customers next quarter at the same rates, the company needs ~240 leads. Improving the proposal win rate from 42% to 55% would deliver the same result with ~180 leads — often a cheaper lever than buying more traffic.

63.3 Customer economics

FORMULA · Core customer metrics

CAC = Sales and marketing cost for new customers ÷ New customers acquired
LTV (simple) = Average gross profit per customer per period ÷ Churn rate per period
LTV/CAC — commonly targeted at ≥ 3 in subscription businesses (heuristic)
CAC payback (months) = CAC ÷ Monthly gross profit per customer
Net revenue retention (NRR) = (Starting recurring revenue + expansion − contraction − churn) ÷ Starting recurring revenue

WORKED EXAMPLE · Unit economics (fictional)

Monthly subscription $400; gross margin 80% → $320 gross profit per month. Monthly churn 2%. Sales and marketing spend $240,000 per quarter; 60 new customers.

- CAC = 240,000 ÷ 60 = **$4,000**.
- LTV = 320 ÷ 0.02 = **$16,000**; LTV/CAC = **4.0**.
- Payback = 4,000 ÷ 320 = **12.5 months**. If churn rises to 3%, LTV falls to $10,667 and LTV/CAC to 2.7 — retention is as important as acquisition.

63.4 Retention

Retention is usually cheaper than acquisition and drives long-term value. Tools:

- **Onboarding:** fast time-to-value; clear success criteria.
- **Customer success / account management:** proactive reviews, usage monitoring, health scores.
- **Service quality:** OTIF, response times, complaint resolution.
- **Commercial structures:** multi-year contracts, volume tiers, loyalty rebates, bundles, switching costs (integration).
- **Voice of the customer:** NPS, satisfaction surveys, churn interviews.

WORKED EXAMPLE · Cohort retention analysis

Cohort (month acquired)	Customers	Active after 3 months	After 6	After 12
January	100	91%	85%	76%
April	120	88%	80%	—
July (discount campaign)	200	72%	58%	—

The July cohort, acquired with deep discounts, churns much faster: the campaign bought low-quality customers. Total customer count hides this; cohort analysis reveals it.

PRACTICE · Practice questions

1. Marketing spend $90,000 and sales salaries/commissions for new business $150,000 bring 80 customers. Average gross profit per customer is $250 per month; monthly churn 2.5%. Compute CAC, LTV, LTV/CAC and payback.
2. Starting ARR $2.0m; expansion $0.35m; contraction $0.10m; churned $0.15m. Compute NRR.
3. A funnel converts leads → opportunities at 20% and opportunities → won at 25%. Average deal $40,000. How many leads are needed for $2m of new business?

ANSWER · Answers to the practice questions

1. CAC = 240,000 ÷ 80 = **$3,000**; LTV = 250 ÷ 0.025 = **$10,000**; LTV/CAC = **3.3**; payback = **12 months**.
2. (2.0 + 0.35 − 0.10 − 0.15) ÷ 2.0 = **105%**.
3. Deals needed = 50; opportunities = 200; leads = **1,000**.

CHAPTER 64

The B2B Sales Process: From Prospect to Signed Agreement

LEARNING OBJECTIVES · In this chapter

A complete B2B sales process — prospecting, qualification, discovery, solution design, proposal, negotiation, closing and handover — with scripts, emails and meeting scenarios.

64.1 The stages

Stage	Goal	Exit criteria	Typical tools
1. Research and targeting	Identify accounts fitting the ICP	Account list with triggers	CRM, industry databases, LinkedIn, trade data
2. Prospecting	Start a conversation	Meeting booked	Calls, emails, referrals, events
3. Qualification	Decide whether to invest effort	Need, budget, authority, timing confirmed	BANT, MEDDICC
4. Discovery	Understand problems, impact, decision process	Documented pains and success criteria	Discovery guide
5. Solution and validation	Design and prove the solution	Demo, pilot, references accepted	Demos, pilots, site visits
6. Proposal	Present offer and business case	Proposal delivered and reviewed	Proposal template, ROI model
7. Negotiation	Agree commercial and legal terms	Agreed term sheet / contract draft	Concession plan, deal desk
8. Closing	Signature	Signed contract, PO	E-signature, approvals
9. Handover and onboarding	Deliver value	Kick-off, implementation plan	Handover document

64.2 Prospecting

DIALOGUE · Cold call opening (fictional; seller of customs-automation software)

Seller: Good morning, Mr. Saleh. This is Rami from ClearPath. I'll be brief — is this a reasonable moment for two minutes? **Prospect:** Two minutes. **Seller:** Thank you. We work with forwarders in Jeddah and Dammam that file more than a thousand declarations a month. Most of them tell us their teams spend hours retyping invoice data and fixing HS-code errors. I'm calling to ask how your team handles that today. **Prospect:** We have six people doing declarations. It's manual but it works. **Seller:** Understood. When volumes peak — say in Ramadan or before year-end — what happens to clearance times? **Prospect:** They slip. We pay overtime and sometimes demurrage. **Seller:** That's exactly what our clients wanted to fix. Would it be worth a 30-minute meeting next week to see whether we could reduce that peak workload? If it isn't relevant, I'll tell you so. **Prospect:** Tuesday morning, then.

Principles: research before contact; respect time; lead with a relevant problem, not product features; ask a question; ask for a small next step.

TRAINING SIMULATION · FICTIONAL DOCUMENT · NOT A VALID INSTRUMENT

Prospecting email (training template)

Subject: Clearance peaks at Dammam — quick question

Dear Mr. Saleh,

Forwarders filing over 1,000 declarations a month in the Eastern Province tell us that peak periods create overtime costs and occasional demurrage, mainly because invoice data is re-keyed manually.

At Gulf Freight (fictional), automating data extraction reduced average preparation time per declaration from 14 to 4 minutes and avoided demurrage on most peak-period containers.

Would a 30-minute call next Tuesday or Wednesday be useful to see whether similar results are realistic for your team? If not a priority now, a short reply would help me not to follow up unnecessarily.

Kind regards, Rami Khoury · ClearPath (fictional) · +966 5X XXX XXXX

64.3 Qualification

Frame-work	Elements
BANT	Budget, Authority, Need, Timing
MEDDICC	Metrics (quantified value), Economic buyer, Decision criteria, Decision process, Identify pain, Champion, Competition

Qualification is about **not wasting resources**: a good salesperson disqualifies early. Signals of poor qualification: no access to decision-makers, no quantified problem, "just collecting quotes", no timeline, procurement-only contact.

64.4 Discovery meetings

PROFESSIONAL INSIGHT · A discovery meeting guide

1. **Agenda and permission:** "May I ask a few questions about how you handle X today, then show only what is relevant?"
2. **Situation:** current process, volumes, systems, team.
3. **Problems:** what goes wrong, how often, who is affected.
4. **Impact:** cost, time, risk, revenue lost — quantify together.
5. **Desired outcome:** what success looks like in 12 months; how measured.
6. **Decision process:** who else is involved, approvals, budget cycle, procurement steps, timing.
7. **Alternatives:** what else they are considering (including doing nothing).
8. **Next step:** agree a concrete action with a date. Listen more than you talk (a common guideline is 70/30). Summarise back what you heard.

DIALOGUE · Discovery: quantifying impact (extract)

Seller: You mentioned demurrage during peaks. Roughly how many containers were delayed last quarter? **Buyer:** Maybe 40. **Seller:** And the average delay and charge? **Buyer:** Two or three days; around 300 riyals a day per container, sometimes more. **Seller:** So approximately 40 × 2.5 days × SAR 300 — about SAR 30,000 last quarter, plus overtime. Is that a fair estimate? **Buyer:** Probably low, if I include customer complaints. **Seller:** Then the problem is worth at least SAR 120,000 a year in direct costs. If we could cut that by two-thirds, would that justify a project this year? **Buyer:** If the numbers

hold, yes — but the general manager decides. **Seller:** Could we include him in the next meeting where we validate these numbers with your data?

64.5 Proposals

A strong proposal is a decision document for the buyer's internal approval, not a brochure. Structure:

1. Executive summary — the client's objective, recommended solution, value, investment, next steps (one page).
2. Understanding of the situation and needs (in the client's words).
3. Proposed solution and scope (what is included and excluded).
4. Implementation plan, timeline and responsibilities.
5. Business case — quantified benefits, costs, payback.
6. Commercial terms — prices, payment terms, validity, assumptions.
7. Why us — relevant references, team, guarantees.
8. Terms and conditions (or reference to master agreement).

A full sample proposal appears in Chapter 66.

64.6 Closing

Closing is the natural outcome of a well-managed process, not a trick. Practical closing techniques:

Technique	Example
Summary close	"We agreed on scope, timeline and price; the remaining step is your signature by the 20th so we can start on 1 March."
Next-step close	"Shall we schedule the kick-off for the week of 4 March?"
Option close	"Would you prefer the 12-month or 24-month agreement?"
Deadline linked to real events	"The installation team is available in March; after that the next slot is May." (Only if true.)
Trial close (checking readiness)	"If we resolve the data-migration question, is there anything else preventing a decision?"

COMMON MISTAKES · Unethical or counterproductive tactics

Fake deadlines, false scarcity, misrepresenting competitors, or promising what operations cannot deliver may win one deal but destroy trust, create disputes and, in some cases, breach consumer-protection or competition law.

64.7 Handover to delivery

Before signature, sales should confirm with operations that the promised scope, timeline and service levels are deliverable. After signature, a handover document records what was sold, promises made, key contacts, risks and success criteria — preventing the classic "sales sold something we cannot deliver".

PRACTICE · Practice questions

1. Write three discovery questions to quantify the impact of late deliveries for a construction-materials buyer.
2. A prospect says: "Send me your price list, we'll compare." How do you respond?
3. Draft a follow-up email after a discovery meeting summarising pains, value and next steps.

ANSWER · Answers to the practice questions

1. “How many deliveries arrived late last quarter, and by how long on average?” “What does a one-day delay cost you on site — idle crews, equipment rental, penalty clauses?” “Have late deliveries led to liquidated damages or disputes with your clients? What was the amount?”
2. Acknowledge and redirect: “Happy to. Prices depend on volumes and service levels, so a list may not compare like with like. Could we spend 20 minutes understanding your requirements — then I can send a quotation you can compare accurately with others?” If they refuse, qualify them as low priority.
3. Example: “Thank you for your time today. My summary: your team files ~1,500 declarations monthly; peak periods cause 2–3 day delays on ~40 containers per quarter (est. SAR 120,000/year in demurrage and overtime); success means peak-period clearance within 48 hours. Next steps: (1) you share last quarter’s demurrage report by Thursday; (2) we prepare a validation with your data; (3) meeting with Mr. Saleh and your GM on the 12th. Please correct anything I misunderstood.”

CHAPTER 65

Pricing, Discounting and Commercial Proposals

LEARNING OBJECTIVES · In this chapter

How professional sellers set prices, govern discounts, structure terms, and write commercial proposals — with a complete annotated proposal.

65.1 Price setting in practice

Prices are set within three boundaries:

- **Floor:** cost (variable cost for incremental deals; full cost for long-run viability).
- **Ceiling:** value to the customer (the next-best alternative's total cost plus the differentiated value you provide).
- **Reference:** competitors' prices and the customer's perception.

Value-based pricing captures part of the value created; **cost-plus** ignores value; **competitor-based** ignores both value and costs unless used carefully.

WORKED EXAMPLE · Economic value to the customer (EVC)

A new pump costs a farm $4,000 vs a standard pump at $3,000. It saves $600 a year in diesel over a 5-year life (discount rate 12% → PV ≈ $2,163) and needs $200 less maintenance per year (PV ≈ $721). EVC = reference price $3,000 + differentiation value $2,884 = **$5,884**. Pricing at $4,000 leaves the farmer $1,884 of value — a compelling offer if the savings are credible and the farmer can finance the higher upfront cost.

65.2 Discount governance

Uncontrolled discounting is one of the largest sources of profit leakage. Good practice:

Control	Description
Price list and floor prices	Approved list; floors by product/segment
Discount authority matrix	Salesperson up to 5%; sales manager 10%; commercial director 15%; CFO beyond
Deal desk	Reviews non-standard deals: price, terms, credit, legal
Give–get rule	Every concession is traded for something: volume, term, prepayment, faster payment, reference, exclusivity
Pocket-price reporting	Measures all deductions (rebates, free freight, terms) by customer
Commission on margin	Aligns salesperson incentives

65.3 Payment terms as part of the price

Extending terms from 30 to 90 days on a $1m order at a 10% cost of capital costs about $16,400 — equivalent to a 1.6% discount. Treat terms requests as price requests.

65.4 Anatomy of a commercial proposal

TRAINING SIMULATION · FICTIONAL DOCUMENT · NOT A VALID INSTRUMENT

Commercial Proposal (training template, abridged)

From: Delta Clear Logistics (fictional), Dammam · **To:** Al-Waha Industrial Supplies (fictional), Riyadh · **Ref.:** DCL-P-2026-044 · **Valid until:** 30 days from issue

1. Executive summary. Al-Waha imports ~1,100 containers a year through Dammam and experiences average clearance of 5.2 days, causing ~SAR 480,000 of annual demurrage, storage and overtime. Delta Clear proposes an integrated customs-clearance and port-trucking service with a 48-hour clearance commitment for compliant shipments. Expected net saving: **SAR 310,000 per year** after our fees, payback immediate. We propose a 2-year agreement starting 1 April.

2. Understanding of your situation. [Volumes, ports, product categories, current providers, issues identified in discovery, success criteria agreed with your logistics manager.]

3. Scope of services. Customs clearance (classification review, declarations, duty payment facilitation); delivery-order collection; port trucking to your Riyadh dry port facility; pre-clearance document checks; monthly KPI report. **Excluded:** customs duties and VAT (paid on your behalf at cost against funding), inspection-authority fees, storage caused by documents not received 72 hours before arrival.

4. Service levels.

KPI	Commitment	Remedy
Clearance within 48 hours of complete documents	95% of containers monthly	Service credit 5% of monthly clearance fees per percentage point below target (max 25%)
Trucking pickup within 24 hours of release	97%	Delta Clear bears demurrage caused by its delay
KPI report	By 5th working day	—

5. Implementation. Weeks 1–2: data mapping and authorisation letters; weeks 3–4: pilot on 50 containers; week 5: full go-live.

6. Pricing (SAR, excluding VAT).

Item	Unit	Price
Customs clearance per declaration	per declaration	450
Port trucking Dammam → Riyadh dry port (40ft)	per container	1,950
Delivery-order handling	per B/L	150
Duty-payment facilitation	% of amount advanced	1.0%

Volume rebate: 3% on clearance fees above 1,000 declarations per year.

7. Business case. Current annual cost SAR 3.35m (clearance, trucking, demurrage, overtime) → projected SAR 3.04m; saving SAR 310,000 (9%).

8. Commercial terms. Payment 30 days from invoice; duty advances reimbursed within 7 days; prices fixed for 12 months, then adjusted by the change in diesel price index for the trucking component; agreement subject to Delta Clear's standard terms of service (attached) and limitation of liability per those terms.

9. Why Delta Clear. Licensed brokerage at all Eastern Province ports; 28 trucks owned; references available from two industrial importers.

10. Next steps. Confirm acceptance by signing the attached order form; kick-off meeting week of 18 March.

Training simulation — fictional companies and figures.

Annotations: the executive summary states the client's objective and the numbers first; the scope lists **exclusions** (where disputes arise); service levels include **remedies**; price adjustment clauses protect against

fuel inflation; payment terms and duty advances are separated (advancing duties is financing and carries credit risk).

PRACTICE · Practice questions

1. A customer requests 120-day terms instead of 30 on $600,000 of annual purchases (evenly spread). Your cost of capital is 9%. What is the annual cost of the request, and what would you ask in return?
2. Your salesperson can approve 5% discounts. A customer asks for 12% for a 3-year contract. Who approves, and what "gets" might justify it?
3. Rewrite a weak proposal opening — "We are a leading provider of innovative solutions" — into a client-focused executive-summary sentence.

ANSWER · Answers to the practice questions

1. Extra receivables ≈ 600,000 × 90/365 = $147,945 → cost ≈ 9% × 147,945 = **$13,315 a year** (≈ 2.2% of sales). Ask for a price increase, a volume commitment, prepayment of part, credit insurance cost-sharing, or supply-chain finance through their bank.
2. Beyond 10% → commercial director (or CFO per the matrix). Gets: 3-year commitment with minimum volumes, prepayment or shorter payment terms, exclusivity in a category, case-study reference, index-linked price review after year 1.
3. "Al-Waha loses an estimated SAR 480,000 a year to clearance delays at Dammam; our proposal reduces clearance to 48 hours and saves about SAR 310,000 a year from the first year."

CHAPTER 66

Negotiation in Practice

LEARNING OBJECTIVES · In this chapter

Principles of commercial negotiation, preparation, tactics and counter-tactics, cultural considerations, and negotiation exercises with model answers.

66.1 Principles

1. **Prepare more than the other side.** Most negotiations are won before the meeting.
2. **Know your BATNA and walk-away point,** and estimate theirs.
3. **Separate interests from positions.** A position is what someone says they want (“10% discount”); an interest is why (“my budget was cut”, “my boss needs to show savings”).
4. **Expand the pie before dividing it:** trade variables valued differently by each side (price, volume, term, payment, delivery, specification, risk allocation, exclusivity).
5. **Make every concession conditional** (“if you…, then we…”), and make concessions smaller as you approach your limit.
6. **Anchor thoughtfully:** a credible first offer influences the outcome.
7. **Keep relationships intact,** especially in repeated business.

66.2 Preparation sheet

Item	Our side	Their side (estimated)
Objectives	Price ≥ $X; 2-year term; 30-day payment	Lower cost; flexibility; less risk
Interests	Margin; volume to fill capacity; reference client	Budget pressure; supply security; quality
BATNA	Sell capacity to segment B at lower margin	Current supplier at +4% price
Walk-away point	$X – 6% with 2-year term	Current supplier’s price
Variables and their value to each side	Term (high for us), prepayment (medium), delivery schedule (low cost to us, high value to them)	
Concession plan	Order, size, conditions	
Questions to ask	What drives your timeline? How do you evaluate suppliers?	
Team roles	Lead, analyst, note-taker; who decides	

66.3 Common tactics and responses

Tactic	What it looks like	Response
Anchoring	“Our budget is 30% below your price.”	Don’t accept the frame; re-anchor with value and your own reference
Nibbling	Small extra demands after agreement	“Happy to look at it as part of the whole package” — trade it
Good cop / bad cop	One friendly, one aggressive	Address the real decision-maker; stay on substance

Tactic	What it looks like	Response
Deadline pressure	"Decide today or we go elsewhere."	Test it; know your BATNA; don't concede for time alone
Higher authority	"My board won't accept this."	Ask what the board needs; ask to meet the decision-maker; reserve your own authority too
Silence	Waiting for you to fill the gap	Stay silent; summarise
Salami (splitting the difference repeatedly)	Small cuts	Concede in decreasing steps; link to conditions
Competitor bluff	"Supplier X offered 15% less."	Ask about scope and terms; compare like for like; hold if confident in value

66.4 Cultural and regional considerations

Negotiation styles vary by culture and individual. In much of the Middle East, relationships and trust built over several meetings, respect for hierarchy, hospitality, and indirect communication are often important; decisions may be made by owners or senior family members rather than the person at the table. Avoid stereotypes — read the individuals, ask about decision processes, and allow time. In tenders and with government entities, formal procedures and compliance rules override relationship-based flexibility.

بالعربية — مبادئ التفاوض التجاري

يعتمد التفاوض الناجح على التحضير الجيد: معرفة البديل الأفضل في حال عدم الاتفاق (BATNA) وحد الانسحاب، وفهم مصالح الطرف الآخر وليس مواقفه المعلنة فقط، وتوسيع نطاق التفاوض ليشمل متغيرات عديدة كالسعر والكمية والمدة وشروط الدفع والتسليم. ويجب ربط كل تنازل بمقابل («إذا قمتم بكذا، نستطيع كذا»)، وتقليل حجم التنازلات كلما اقتربنا من حدنا الأدنى، والحفاظ على العلاقة لأن معظم التعاملات التجارية متكررة.

66.5 Negotiation exercises

PRACTICE · Scenario 1 — supplier price increase

You are the procurement manager of a food manufacturer. Your packaging supplier announces a 12% price increase citing resin costs. Annual spend $2.4m. Resin indices rose 15% over 6 months; resin is ~55% of the supplier's cost. You have one alternative supplier, 6% more expensive than current prices, needing 3 months to qualify. Prepare your position.

ANSWER · Model approach for Scenario 1

- Justified increase ≈ 55% × 15% ≈ **8.3%**, not 12%.
- BATNA: alternative at +6% after qualification (plus switching costs) — so a settlement between 6% and 8% is realistic.
- Proposal: accept an **index-linked formula** (55% of price moves with a published resin index, quarterly, with a band), apply it retroactively from now (≈ +8%), in exchange for a 2-year agreement, 60-day payment terms and priority supply. Start qualifying the alternative anyway to strengthen your BATNA.

PRACTICE · Scenario 2 — distributor asks for exclusivity

A European equipment maker (you) is approached by a Gulf distributor asking for 5-year exclusivity in all GCC countries. The distributor has strong relationships in Saudi Arabia but none in Qatar or Oman. What do you offer?

ANSWER · Model approach for Scenario 2

Offer exclusivity in Saudi Arabia (where it can perform) for an initial 2–3 years, renewable, conditional on **minimum purchase targets**, marketing and service commitments, stock levels, trained technicians, and reporting; non-exclusive rights elsewhere until performance is proven. Include termination rights for missed targets, clear post-termination stock buy-back terms, and check local agency/distribution laws (some jurisdictions give registered agents protection on termination — legal advice needed).

PRACTICE · Scenario 3 — customer demands 5% discount at renewal

A logistics client representing 8% of your revenue says a competitor offers 5% lower rates. Your gross margin on the account is 14%. What do you do?

ANSWER · Model approach for Scenario 3

A 5% price cut would reduce gross profit on the account by about a third (5 ÷ 14 ≈ 36%). Ask for the competitor's offer scope (service levels, free time, liability, payment terms). Present your performance data (OTIF, claims ratio, avoided demurrage). Offer alternatives: a 2% rebate for a 2-year extension and volume commitment, efficiency changes that lower your cost (consolidated pickups, digital documents), or adjust service levels. Decide in advance your walk-away: losing the account may be better than serving it at near-zero margin.

CHAPTER 67

Account Management, CRM, Sales Forecasting and Revenue Operations

LEARNING OBJECTIVES · In this chapter

Managing and growing existing customers, key account management, CRM systems and data, sales forecasting methods, and the revenue operations function.

67.1 Key account management (KAM)

Key accounts are selected for strategic importance, not just size: revenue and profit potential, growth, reference value, strategic fit. KAM practices:

- **Account plan:** customer's strategy and priorities, our share of wallet, growth opportunities, risks, relationship map, actions, targets.
- **Relationship mapping:** multiple contacts at multiple levels ("multi-threading"), not a single point of contact.
- **Joint business reviews (QBRs):** performance against KPIs, issues, improvement plans, upcoming needs.
- **Value tracking:** document savings or benefits delivered — used at renewal.
- **Risk monitoring:** credit exposure, concentration, competitor activity, contract expiry.

67.2 CRM systems

A CRM records accounts, contacts, activities, opportunities, quotes, contracts and service cases. Value comes from **data discipline**: opportunities with correct stages, values and close dates; activities logged; one record per account. CRM data feeds pipeline reports, forecasts and incentive calculations; poor data produces poor forecasts.

67.3 Sales forecasting

Method	Description	Strengths	Weaknesses
Weighted pipeline	Σ opportunity value × stage probability	Simple, systematic	Probabilities generic; stale deals inflate
Commit / best case / pipeline	Salespeople categorise deals	Uses judgement	Optimism or sandbagging
Historical run-rate / time series	Extrapolate past patterns	Good for transactional, recurring sales	Misses structural change
Driver-based	Leads × conversion × deal size; customers × usage	Links to activity	Needs reliable conversion data
Cohort-based (recurring revenue)	Retained revenue + new bookings	Fits subscriptions	Needs cohort data

WORKED EXAMPLE · Weighted pipeline forecast (fictional quarter)

Stage	Prob-ability	Pipeline value	Weighted
Qualified	10%	$2.0m	$0.20m
Discovery complete	25%	$1.6m	$0.40m
Proposal	50%	$1.0m	$0.50m
Negotiation	75%	$0.8m	$0.60m
Verbal commit	90%	$0.4m	$0.36m
Total		**$5.8m**	**$2.06m**
Target $2.4m → gap $0.34m. Actions: accelerate the negotiation-stage deals, add pipeline, or adjust the plan. Historical accuracy should be checked: if the team usually closes only 80% of weighted pipeline, forecast ≈ $1.65m.			

67.4 Revenue operations (RevOps)

RevOps aligns marketing, sales and customer success operations: shared definitions (what is an SQL?), systems (CRM, marketing automation, CPQ — configure, price, quote), data quality, territory and quota design, incentive compensation administration, forecasting process and analytics. For a finance–CS professional, RevOps is a natural bridge role.

PRACTICE · Practice questions

1. Build a weighted forecast: opportunities of $300k (proposal, 50%), $500k (discovery, 25%), $200k (negotiation, 75%), $150k (verbal, 90%). Compute the total.
2. List five elements of a key account plan.
3. Why do salespeople sometimes “sandbag” forecasts, and how can management reduce it?

ANSWER · Answers to the practice questions

1. 150 + 125 + 150 + 135 = **$560k**.
2. Customer objectives and strategy; current revenue, margin and share of wallet; relationship map; growth opportunities with values and timing; risks and mitigation; action plan with owners and dates; KPIs and review schedule.
3. To beat targets and protect bonuses (low forecasts are easier to exceed). Reduce it with forecast-accuracy metrics, incentive designs that reward accurate forecasting, deal inspection based on evidence (buyer actions, not seller feelings), and data-driven stage probabilities.

CHAPTER 68

Channels and Go-to-Market: B2B, B2C, Enterprise, Distributors, Agents, Resellers and Partners

LEARNING OBJECTIVES · In this chapter

Differences between B2B, B2C and enterprise selling; indirect channels and their agreements; how to choose and manage channel partners.

68.1 B2B vs B2C vs enterprise

Dimension	B2C	SMB B2B	Enterprise B2B
Decision-makers	Individual/household	Owner or small team	Buying committee (5–15 people), procurement, legal
Sales cycle	Minutes to days	Weeks	Months to over a year
Deal size	Small	Small–medium	Large
Channel	Retail, e-commerce, apps	Inside sales, partners, online	Field sales, key account teams, partners
Pricing	List prices, promotions	Standard tiers	Negotiated, custom contracts
Contract	Terms of sale (consumer law)	Standard terms	Master agreements, SLAs, security reviews
Key metrics	Conversion, basket, retention	CAC, churn	Pipeline coverage, win rate, NRR

68.2 Indirect channels

Channel partner	Role	Owns goods?	Earns	Key agreement terms
Agent (commercial agent)	Finds customers; may negotiate/contract for principal	No	Commission	Territory, commission, exclusivity, termination compensation (some laws)
Distributor	Buys, stocks, resells, often provides service	Yes	Margin	Territory, exclusivity, targets, pricing freedom (competition law), stock, service, IP use, termination and buy-back
Reseller / VAR	Resells (often software/IT), may add services	Usually	Margin or discount	Partner tier, certifications, deal registration
Wholesaler	Buys in bulk, sells to retailers	Yes	Margin	Volumes, terms
Retailer	Sells to consumers	Yes (or consignment)	Margin	Listing fees, promotions, payment terms
Marketplace / platform	Connects buyers and sellers	Usually not	Take rate	Fees, rules, data
Referral partner / alliance	Introduces opportunities	No	Referral fee or reciprocity	Fee terms, confidentiality

Channel partner	Role	Owns goods?	Earns	Key agreement terms
Franchisee	Operates under brand and system	Owns its business	Profit minus royalties	Franchise agreement (Part XIII)

COMMON MISTAKES · Legal points to check for agents and distributors

Many countries have specific laws on commercial agency and distribution — registration requirements, restrictions on termination, compensation for agents on termination (e.g., under EU-derived laws), exclusive registered agent regimes in some Middle Eastern jurisdictions, and competition-law limits on resale price maintenance and territorial restrictions. Use local legal advice before appointing or terminating.

68.3 Choosing direct or indirect

Factor	Favours direct sales	Favours partners
Deal size	Large, complex	Small, standard
Need for local presence/service	Can afford own team	Partner already present
Customer relationship and data	Critical to own	Less critical
Speed of market entry	Slower	Faster
Control of pricing and brand	High	Lower
Cost structure	Fixed (salaries)	Variable (margin/commission)

68.4 Managing channel partners

Partner selection (market coverage, capabilities, financial strength, reputation, compliance — anti-bribery due diligence is essential where partners deal with government buyers), onboarding (training, materials), joint business plans with targets, incentives (rebates, MDF — market development funds), performance reviews, channel conflict management (deal registration, territory rules), and exit planning.

PRACTICE · Practice questions

1. A software company sells a $3,000/year product to SMEs in 12 Arab countries. Direct or partner model? Explain.
2. A distributor buys at 30% below list and sells at list. An agent alternative would earn 8% commission and the company would sell directly at list. Compare the company's revenue per $100 of list-price sales and the risks it carries under each.
3. Name three clauses you would insist on in a distribution agreement as the supplier.

ANSWER · Answers to the practice questions

1. Partners (resellers/MSPs) or a digital self-serve model: deal size too small to justify field sales in 12 markets; partners provide local language support, invoicing and relationships. Possibly direct digital sales with local partners for implementation.
2. Distributor: company revenue $70; distributor holds stock, credit risk on end customers and local marketing. Agent: company revenue $100 minus $8 commission = $92 net, but the company bears credit risk, stock, logistics, invoicing and possibly local tax registration. The extra $22 must cover these costs and risks.
3. Minimum purchase/performance targets with termination right; territory and exclusivity scope (and limits consistent with competition law); IP/trademark use and quality standards;

confidentiality; payment terms and security (e.g., SBLC); stock buy-back and transition on termination; anti-bribery and sanctions compliance; governing law and arbitration.

SUMMARY · Part VIII summary

- Commercial success starts with choosing the right customers and a quantified value proposition.
- Funnel metrics and customer economics (CAC, LTV, retention) show whether growth creates value.
- B2B selling is a disciplined process: qualify hard, discover deeply, propose for the buyer's internal decision, and close on clear next steps.
- Price is the strongest profit lever; govern discounts and treat payment terms as price.
- Negotiation is preparation: BATNA, interests, variables and conditional concessions.
- Key accounts need plans and multi-level relationships; forecasts need clean CRM data.
- Channels extend reach at the cost of margin and control; agreements must reflect local law.

PART IX

Procurement, Logistics and Supply Chain

المشتريات والخدمات اللوجستية وسلاسل الإمداد

If sales is how a business earns money, procurement and supply chain are how it spends most of it. For a distributor, purchases can be 70–85% of revenue; a 2% saving in landed cost can be worth more than a 20% increase in sales. Supply chains also determine service levels, working capital and resilience to disruption.

This part covers the full chain: procurement strategy and supplier management, inventory and warehousing, transport modes and freight economics, customs and disruption management, and a complete simulation of importing goods from China into the Middle East with every cost from supplier quotation to the buyer's warehouse.

LEARNING OBJECTIVES · **Learning objectives for Part IX**

- Design a procurement strategy and run RFI, RFQ and RFP processes with objective evaluation.
- Qualify suppliers, negotiate and contract, and choose payment terms and supply-chain finance.
- Manage inventory with forecasts, reorder points, safety stock and turnover targets.
- Compare transport modes and price freight correctly (chargeable weight, W/M, surcharges).
- Understand ports, container and bulk shipping, multimodal transport and last-mile delivery.
- Manage customs, demurrage and detention, and supply-chain disruption.
- Calculate a full landed cost and allocate costs across products.

Chapters in this part

Chapter 69 Procurement Strategy and Supplier Management 29
Chapter 70 Inventory Management and Warehousing 33
Chapter 71 Transport Modes, Freight and Logistics Networks 36
Chapter 72 Customs, Demurrage, Supply-Chain Finance and Disruption Management 39
Chapter 73 Simulation: Importing from China into the Middle East 42

CHAPTER 69

Procurement Strategy and Supplier Management

LEARNING OBJECTIVES · In this chapter

Strategic sourcing, category segmentation, supplier identification and qualification, RFI/RFQ/RFP processes, evaluation, negotiation, contracts, payment terms and supplier performance management.

69.1 From purchasing to strategic procurement

Level	Focus	Question
Transactional purchasing	Processing orders	"Did we get it on time?"
Tactical procurement	Price negotiation, supplier choice	"Did we get a good price?"
Strategic sourcing	Total cost, supply risk, innovation, category strategies	"Are we buying the right things from the right suppliers on the right terms, with acceptable risk?"

69.2 Category segmentation: the Kraljic matrix

	Low supply risk	High supply risk
High profit impact	**Leverage items** — many suppliers, high spend (e.g., packaging, standard steel). Strategy: competitive tendering, volume consolidation, negotiate hard.	**Strategic items** — few suppliers, critical (e.g., key components, specialised chemicals). Strategy: partnerships, long-term contracts, joint development, risk mitigation.
Low profit impact	**Non-critical items** — office supplies, MRO. Strategy: simplify, catalogues, e-procurement, reduce transaction cost.	**Bottleneck items** — low spend but scarce (e.g., a proprietary spare part). Strategy: secure supply, safety stock, find alternatives, redesign.

69.3 Supplier identification and qualification

Sources: trade fairs, industry associations, B2B platforms, trade data (importers' records in some countries), embassies and trade-promotion agencies, referrals, existing suppliers' suppliers.

Qualification checks:

Area	Evidence
Legal existence and ownership	Registration documents, UBO, sanctions and adverse-media screening
Financial strength	Financial statements, credit reports, bank references
Capability and capacity	Factory audit, production lines, capacity utilisation, lead times
Quality	ISO 9001 or sector standards, quality-control process, defect rates, samples, test reports
Compliance	Product certifications, export licences, labour and environmental standards, anti-bribery
Logistics	Proximity to ports, packaging, export experience, Incoterms capability
References	Other customers, ideally in your region

Area	Evidence
Risk	Single-site dependency, country risk, concentration of their revenue on you

69.4 RFI, RFQ and RFP

Document	Purpose	When	Output
RFI — request for information	Learn about suppliers' capabilities	Early, market scanning	Shortlist
RFQ — request for quotation	Get prices for well-specified items	Standard goods/services	Price comparison
RFP — request for proposal	Get solutions and prices for complex needs	Services, systems, projects	Evaluated proposals
Tender (often public)	Formal competitive process with strict rules	Government, large corporates	Contract award

TRAINING SIMULATION · FICTIONAL DOCUMENT · NOT A VALID INSTRUMENT

Request for Quotation (training template, abridged)

Buyer: Al-Waha Industrial Supplies (fictional) · **RFQ no.:** AW-RFQ-0312 · **Issue date:** 3 March · **Closing date:** 17 March, 17:00 (Riyadh)

1. **Items:** (a) Galvanised steel pipe, ASTM A53 Grade B, 2", schedule 40, 6 m lengths — 120 tonnes; (b) fittings per attached list.
2. **Delivery:** CFR Dammam, Incoterms® 2020; shipment in two lots: May and July.
3. **Required in quotation:** unit price (USD/tonne), total, validity (min. 30 days), lead time, packing, payment terms, mill test certificates, country of origin, HS code, inspection arrangements, warranty.
4. **Payment terms preferred:** 10% advance against advance-payment guarantee; 90% by L/C at 60 days after B/L date.
5. **Evaluation criteria:** total landed cost (50%), quality and certification (20%), delivery reliability (15%), payment terms (10%), supplier financial strength (5%).
6. **Conditions:** quotations must comply with Saudi product conformity requirements; buyer may award partially; anti-bribery declaration attached; questions by email by 10 March.

Training simulation — fictional.

69.5 Evaluating offers: the procurement comparison sheet

WORKED EXAMPLE · Comparison of three supplier offers (fictional, 120 t steel pipe, CFR Dammam)

Criterion	Supplier A (China)	Supplier B (Turkey)	Supplier C (India)
Price CFR (USD/t)	820	905	845
Duty and charges (assume 5% duty + $35/t)	76	80	77
Payment terms	30% advance T/T, 70% at sight L/C	10% advance, 90% L/C 60 days	100% L/C at sight
Financing cost/benefit vs base (8% p.a.)	Advance 30% held ~75 days: +$4/t	60-day credit on 90%: −$11/t	0
Expected quality cost (rejects, rework)	$30/t (history 3.5% rejects incl. replacement costs)	$5/t	$12/t
Lead time	60 days	35 days	50 days

Criterion	Supplier A (China)	Supplier B (Turkey)	Supplier C (India)
Safety stock implication	+$9/t holding cost	0	+$4/t
Total landed cost (USD/t)	**939**	**979**	**938**
Quality/certification score (0–10)	6	9	7
Delivery reliability score	6	9	7

Weighted scoring (landed cost 50% scored relative to lowest; others as weights): C ranks first narrowly, B second (premium for quality and speed), A third despite the lowest price. Decision: award 70% to C, 30% to B (dual sourcing reduces risk and keeps competition).

69.6 Supplier negotiations and contracts

Negotiate the **total cost**: price, payment terms, Incoterms, packaging, lead time, quality guarantees, warranties and remedies, price-adjustment formulas, minimum order quantities, consignment or VMI stock, rebates, and exclusivity.

Key supply-contract clauses: specifications and acceptance tests; delivery terms and liquidated damages for delay; warranties and remedies; inspection rights; price-adjustment mechanisms (e.g., linked to steel or resin indices); force majeure; intellectual property; confidentiality; compliance (anti-bribery, sanctions, labour standards); limitation of liability; termination; governing law and dispute resolution.

69.7 Supplier payment terms and supply-chain finance

Term	Buyer cash impact	Supplier risk	When used
Advance payment	Worst	None	New supplier relationships, custom goods
Cash against documents	Neutral	Low	Trade with moderate trust
L/C at sight / usance	Neutral / financing	Low	International trade
Open account 30–120 days	Best	High	Established relationships
Consignment	Best (pay when used)	Highest	Strategic partnerships

Supply-chain finance (payables finance) allows suppliers to be paid early by a bank at the buyer's credit cost while the buyer keeps or extends its terms (see Chapter 50 (Volume 4)).

69.8 Supplier performance management

KPIs: on-time-in-full (OTIF), quality (defect rate, PPM), lead time and variability, responsiveness, price competitiveness, compliance, innovation. Supplier scorecards reviewed quarterly; development plans for strategic suppliers; exit plans for chronic underperformers.

PRACTICE · Practice questions

1. Classify in the Kraljic matrix for a dairy processor: (a) milk cartons; (b) office stationery; (c) a specialised cultures supplier with a patent; (d) electricity.
2. A supplier offers a 2% discount if you pay 30% in advance 60 days before shipment and the 70% balance on delivery (delivery is 30 days after shipment), instead of open account 60 days after delivery with no discount. Purchases $1.2m; your cost of capital 9%. Compare.
3. Write four qualification questions for a new overseas supplier of electrical cables.

ANSWER · Answers to the practice questions

1. (a) Leverage (high spend, many suppliers); (b) non-critical; (c) strategic (high impact, limited supply — or bottleneck if spend is small); (d) leverage or strategic depending on market — often strategic with limited supplier choice and high impact.
2. Discount saves **$24,000**. Cost of paying earlier: the 30% advance ($360k) is paid 150 days earlier than under open account (60 + 30 + 60): 360,000 × 9% × 150/365 = $13,315; the 70% balance ($840k) is paid 60 days earlier: 840,000 × 9% × 60/365 = $12,427. Total **$25,742** > $24,000 → slightly unfavourable even before the counterparty risk of the advance. Decline, or accept only with a higher discount or an advance-payment guarantee.
3. Do you hold certifications to IEC/local standards for these cables, and can you provide test reports from accredited labs? What is your annual capacity and current utilisation? Who are three customers in our region we can contact? Can you provide audited financial statements and a bank reference? (Also: copper price adjustment policy, export experience, ownership.)

CHAPTER 70

Inventory Management and Warehousing

LEARNING OBJECTIVES · In this chapter

Demand forecasting, inventory policies, reorder points and safety stock, EOQ, ABC/XYZ analysis, inventory KPIs, and warehouse operations and costs.

70.1 Why inventory exists

Type	Purpose
Cycle stock	Ordering in batches (economies of ordering/shipping)
Safety stock	Protection against demand and supply variability
Pipeline (in-transit) stock	Goods moving through the chain — owned by whoever bears it under the Incoterms/contract
Anticipation / seasonal stock	Build-up before peaks (Ramadan, summer, back-to-school)
Speculative stock	Buying ahead of expected price rises (a commercial bet — label it as such)
Decoupling stock	Buffers between production stages

70.2 Demand forecasting

Methods range from moving averages and exponential smoothing to regression and machine learning (Part XVI). Forecasts should be made at the level decisions are taken (SKU × location × week), reconciled with sales and marketing plans in a **Sales and Operations Planning (S&OP)** process, and measured for accuracy (MAPE) and bias.

70.3 Reorder points, safety stock and service levels

FORMULA · Reorder point and safety stock

ROP = average daily demand × average lead time (days) + safety stock
Safety stock (demand variability only) = $z \times \sigma_d \times \sqrt{L}$
Safety stock (demand and lead-time variability) = $z \times \sqrt{(L \times \sigma_d^2 + d^2 \times \sigma_L^2)}$
z = service-level factor (90% → 1.28; 95% → 1.65; 98% → 2.05; 99% → 2.33); σ_d = std dev of daily demand; L = lead time in days; σ_L = std dev of lead time

WORKED EXAMPLE · Air-conditioner spare parts (fictional)

Average demand d = 40 units/day; σ_d = 12; lead time L = 30 days; σ_L = 6 days; target service level 95% (z = 1.65).

- SS = 1.65 × √(30 × 144 + 1,600 × 36) = 1.65 × √(4,320 + 57,600) = 1.65 × 248.8 ≈ **411 units**.
- ROP = 40 × 30 + 411 = **1,611 units**.
- Lead-time variability (σ_L) dominates: reducing it from 6 to 3 days cuts SS to about 1.65 × √(4,320 + 14,400) = **226 units** — supplier reliability is worth money.

70.4 Order quantity

EOQ (see Part IV) balances ordering and holding costs; in international trade, **container economics** often determine quantities (ordering a full 40ft container may be cheaper per unit than LCL), along with supplier MOQs and price breaks.

70.5 ABC and XYZ analysis

- **ABC** by value: A items (~10–20% of SKUs, ~70–80% of value) — tight control, frequent review; C items (~50% of SKUs, ~5% of value) — simple rules.
- **XYZ** by demand variability: X stable, Y variable, Z erratic. AX items suit automated replenishment; CZ items may be made-to-order or dropped.

70.6 Inventory KPIs

KPI	Formula	Use
Inventory turnover	COGS ÷ average inventory	Efficiency
Days inventory outstanding	365 ÷ turnover	Working capital
Fill rate / service level	Orders or lines fully served ÷ total	Customer service
Stock-out rate	Stock-out occurrences ÷ SKUs or orders	Lost sales risk
Obsolete / slow-moving stock %	Value aged > x months ÷ total	Write-off risk
GMROI	Gross margin ÷ average inventory at cost	Retail profitability of stock
Inventory accuracy	Matching system vs physical counts	Data quality

70.7 Warehousing

Functions: receiving, inspection, put-away, storage, picking, packing, dispatch, returns, value-added services (labelling, kitting). Cost drivers: space (per pallet position or m²), labour (per pick, per hour), equipment, systems (WMS), utilities (especially cold storage), insurance. Options: own warehouse (control, fixed cost), leased, public/shared warehouse, 3PL contract (variable cost, flexibility), bonded/free-zone warehouses (duty deferral).

WORKED EXAMPLE · Cost per pallet in a 3PL warehouse (fictional monthly)

Activity	Rate	Volume	Cost
Storage per pallet per week	$3.50	800 pallets × 4.3 weeks	$12,040
Inbound handling per pallet	$4.00	300	$1,200
Outbound picking per case	$0.45	18,000	$8,100
Value-added labelling per unit	$0.08	25,000	$2,000
WMS fee	—	—	$500
Total			**$23,840**
Cost per case shipped ≈ $1.32; as a % of $1.5m monthly COGS shipped ≈ 1.6%.			

PRACTICE · Practice questions

1. Demand 120/day, σ_d 30, lead time 20 days (fixed). Compute safety stock and ROP at 98% service.

2. Inventory: average $4.5m; COGS $27m; gross margin $9m. Compute turnover, DIO and GMROI.
3. Why might a distributor deliberately hold more inventory than EOQ suggests?

ANSWER · Answers to the practice questions

1. SS = 2.05 × 30 × √20 = 2.05 × 134.2 ≈ **275 units**; ROP = 120 × 20 + 275 = **2,675 units**.
2. Turnover = 27 ÷ 4.5 = **6.0**×; DIO ≈ **61 days**; GMROI = 9 ÷ 4.5 = **2.0** ($2 of gross margin per $1 of inventory per year).
3. Supplier MOQs or full-container economics, expected price increases, supply risk (disruptions), seasonal peaks, customer service commitments with penalties, or to secure availability during shortages — each with an explicit cost-benefit justification.

CHAPTER 71

Transport Modes, Freight and Logistics Networks

LEARNING OBJECTIVES · In this chapter

Road, rail, air and sea transport; container and bulk shipping; multimodal transport; ports and terminals; last-mile delivery; freight forwarders; and how freight is priced.

71.1 Comparing modes

Mode	Cost per tonne-km	Speed	Reliability	Suited to	Key documents
Sea (container)	Very low	Slow (2–6 weeks intercontinental)	Moderate (schedule reliability varies)	Most manufactured goods	B/L, sea waybill
Sea (bulk/tanker)	Lowest	Slow	Depends on charter	Commodities: grain, ore, coal, oil	B/L, charter party
Air	Highest	Fastest (days)	High	High value, urgent, perishable, small	Air waybill
Road	Moderate	Door-to-door flexibility	Border delays possible	Regional trade (e.g., GCC, Levant), last mile	CMR note, TIR carnet
Rail	Low–moderate	Medium	Good on corridors	Bulk and containers on long land corridors	CIM/SMGS consignment notes
Pipeline	Very low (after construction)	Continuous	High	Oil, gas, water	Contracts
Multimodal	Combination	—	One contract for the whole route	Door-to-door	Multimodal/combined transport B/L

71.2 Container shipping

- **Equipment:** 20ft (1 TEU, ~33 m³), 40ft (2 TEU, ~67 m³), 40ft high cube (~76 m³), reefers (refrigerated), open-top, flat-rack, tank containers.
- **FCL vs LCL:** full container load (one shipper) vs less-than-container load (consolidated by a forwarder/NVOCC at a container freight station).
- **Liner services:** scheduled routes; carriers often operate in alliances sharing vessel space.
- **Freight components:** base ocean freight + surcharges: bunker (fuel) adjustment factor (BAF), currency adjustment factor (CAF), peak-season surcharge (PSS), emergency or war-risk surcharges, low-sulphur fuel surcharges, carbon-related surcharges (e.g., EU ETS costs passed through), terminal handling charges (THC) at origin and destination, documentation fees, equipment imbalance charges.
- **Free time:** days without demurrage/detention at origin and destination — negotiable.
- **Verified gross mass (VGM):** shippers must declare container weight before loading (SOLAS requirement since 2016).

71.3 Bulk shipping

Dry bulk (grain, ore, coal, fertiliser) and liquid bulk (crude, products, chemicals, LNG) move under **charter parties**: voyage charter (pay per tonne for a voyage), time charter (hire the vessel per day), bare-

boat charter (lease the vessel without crew). Vessel sizes (e.g., Handysize, Supramax, Panamax, Capesize for dry bulk; VLCC, Suezmax, Aframax for tankers) affect freight per tonne and port access. Laytime, demurrage and despatch are key commercial terms (Part XI).

71.4 Air cargo

Priced per chargeable kg (higher of actual and volumetric weight at 6,000 cm^3/kg), with fuel and security surcharges, airline and handling charges. Air cargo moves in passenger aircraft bellies and freighters. Dangerous goods (including lithium batteries) face strict IATA rules.

71.5 Road and rail

Road freight is priced per trip, per kg or per pallet; cross-border road transport depends on border-crossing times, permits and transit regimes (e.g., TIR carnets). Rail is efficient for long corridors and bulk; new rail links in the Gulf region (e.g., the Etihad Rail network in the UAE) are changing regional logistics options.

71.6 Ports and terminals

A port call involves pilotage, berthing, stevedoring (loading/unloading), terminal storage, customs inspection, and gate-out. Port performance indicators: vessel waiting time, crane productivity, truck turnaround time, dwell time of containers. Major regional hubs include Jebel Ali, King Abdullah Port, Jeddah Islamic Port, Dammam, Salalah, Hamad Port, Port Said and others. Port congestion and dwell times directly affect demurrage and inventory planning.

71.7 Last-mile delivery

The final delivery to the customer is often the most expensive part per unit (failed deliveries, address quality, cash on delivery in some markets, returns). Solutions: route optimisation software, pickup points/lockers, delivery windows, crowdsourced fleets, address standardisation (e.g., national addressing systems).

71.8 Freight forwarders and NVOCCs

Forwarders organise transport, consolidate cargo, handle documents and customs, and buy capacity from carriers. An **NVOCC** (non-vessel operating common carrier) issues its own house B/Ls and acts as carrier towards the shipper. Choosing a forwarder: network coverage, carrier contracts, customs capability, technology (tracking, documents), financial stability, insurance and liability terms (standard trading conditions often limit liability heavily).

71.9 Freight pricing: worked examples

WORKED EXAMPLE · Container quote (fictional, Shanghai → Jebel Ali, 40ft HC)

Charge	USD
Ocean freight (base)	1,450
Bunker surcharge	320
Origin THC	160
Documentation	60

Charge	USD
Seal and VGM	25
Destination THC	250
Delivery order fee	90
Total port-to-port + terminals	**2,355**
Under FOB Shanghai the buyer pays everything except origin THC (usually the seller's); under CFR Jebel Ali the seller pays freight and surcharges, and the buyer pays destination THC and DO fees (check the carriage contract).	

WORKED EXAMPLE · FCL vs LCL break-even

LCL rate $70 per W/M plus $180 fixed origin/destination CFS charges; FCL 20ft all-in $1,300. LCL cost for x m³ = 70x + 180. Break-even: 70x + 180 = 1,300 → x = **16 m³**. Above ~16 m³, a 20ft container is cheaper (and handling risk is lower).

PRACTICE · Practice questions

1. An air shipment: 8 cartons of 80 × 60 × 50 cm, 22 kg each, rate $3.80/kg. Compute the freight.
2. Why would a shipper of 30 m³ choose a 40ft container over a 20ft?
3. List five surcharges that can appear on a container freight invoice and explain one.

ANSWER · Answers to the practice questions

1. Volume per carton 240,000 cm³ ÷ 6,000 = 40 kg volumetric > 22 kg actual → chargeable 8 × 40 = 320 kg → **$1,216**.
2. A 20ft holds ~33 m³ nominal but ~25–28 m³ usable depending on cargo shape; 30 m³ will not fit comfortably. Also 40ft rates are often well below twice the 20ft rate.
3. BAF (fuel), CAF (currency), PSS (peak season), THC (terminal handling), war-risk/emergency surcharges, low-sulphur surcharge, EU ETS surcharge, equipment imbalance. Example: BAF adjusts freight for changes in bunker fuel prices, usually recalculated monthly or quarterly.

CHAPTER 72

Customs, Demurrage, Supply-Chain Finance and Disruption Management

LEARNING OBJECTIVES · In this chapter

The operational side of customs clearance, how Incoterms allocate costs and risks in supply-chain planning, managing demurrage and detention, using payment terms and supply-chain finance, and building resilience against disruption.

72.1 Customs clearance in operations

The legal and valuation principles of customs are in Chapter 56 (Volume 4). Operationally, clearance succeeds when:

1. **Documents arrive before the goods** (commercial invoice, packing list, B/L or delivery order, certificate of origin, conformity certificates, permits).
2. **Data are consistent** across documents and with the declaration.
3. **HS classification** has been agreed in advance (ideally with a binding ruling for recurring items).
4. **Duties and VAT are funded** (deposit accounts or guarantees with customs, broker advances).
5. **Product compliance** is completed before shipment (e.g., product registration and conformity certificates in the destination country).
6. The **broker** has authorisation (power of attorney / delegation in the customs system).

Many countries use **single-window** systems and risk-based inspection channels (green: release; yellow: document check; red: physical inspection). Trusted-trader (AEO) programmes reduce inspections.

72.2 Incoterms and cost allocation in supply-chain planning

The procurement team must translate the agreed Incoterms rule into a **cost and responsibility plan**:

Responsibility	FOB origin port	CFR destination port	DAP buyer warehouse
Book main carriage	Buyer (via its forwarder)	Seller	Seller
Insurance	Buyer	Buyer (not required of seller)	Seller (bears risk)
Destination THC and DO	Buyer	Buyer (unless included)	Seller
Import clearance and duties	Buyer	Buyer	Buyer
Inland delivery	Buyer	Buyer	Seller
Who controls timing and carrier choice	Buyer	Seller	Seller
Who bears freight-rate changes after contract	Buyer	Seller	Seller

Many experienced importers prefer **FOB/FCA** so that they control carriers, negotiate freight with their volume, and see costs transparently; small importers without logistics capability may prefer **CFR/CIF** or **DAP** from reliable suppliers.

72.3 Managing demurrage and detention (D&D)

Cause	Prevention
Late original documents	Use electronic B/L or telex release once payment is secured; courier documents promptly; present L/C documents early
Customs holds	Pre-clearance checks, correct classification, certificates ready
Warehouse not ready / no trucks	Plan receiving capacity; book trucks in advance
Carrier free time too short	Negotiate extended free time in the service contract (e.g., 14 days combined)
Port congestion	Monitor; route via alternative ports where possible
Unclear responsibility	Contract clauses allocating D&D caused by each party

72.4 Supplier payment terms and supply-chain finance

Payment terms shift working capital between buyers and suppliers. Large buyers extending payment terms can push weak suppliers into distress — which then becomes the buyer's supply risk. **Supply-chain finance** lets suppliers get paid early at a cost based on the buyer's credit, balancing the buyer's working capital with suppliers' liquidity. Other tools: inventory financing, consignment stock, vendor-managed inventory, and dynamic discounting.

72.5 Disruption management and resilience

Disruption type	Examples	Mitigation
Supplier failure	Bankruptcy, fire, quality failure	Dual sourcing, supplier financial monitoring, safety stock
Logistics disruption	Port congestion, canal or strait closures, carrier failures	Alternative routes and ports, buffer lead times, multiple carriers
Geopolitical	Sanctions, conflict, export bans	Screening, diversification of origins, contract clauses
Natural events and pandemics	Floods, earthquakes, epidemics	BCP, geographic diversification
Demand shocks	Sudden spikes or collapses	Flexible contracts, S&OP, scenario planning
Cyber attacks	Carrier or port systems down	Manual fallbacks, cyber requirements for partners
Price shocks	Freight rate spikes, commodity spikes	Indexed contracts, hedging, long-term freight contracts

PROFESSIONAL INSIGHT · Resilience costs money — decide deliberately

Each mitigation has a cost: dual sourcing loses volume discounts; safety stock ties up cash; long-term freight contracts may be above spot in soft markets. A structured approach — map critical items and their single points of failure, estimate impact (lost margin per week of disruption), and buy resilience where the expected loss exceeds the cost — avoids both fragility and waste.

بالعربية — إدارة الاضطرابات في سلاسل الإمداد

أثبتت الأزمات الأخيرة، مثل الجائحة وازدحام الموانئ والاضطرابات في البحر الأحمر، أن سلاسل الإمداد الأرخص ليست بالضرورة الأكثر أماناً. لذلك تحدد الشركات المحترفة المواد الحرجة ونقاط الضعف المنفردة (مورد واحد أو ميناء واحد أو مسار واحد)، وتقدّر

الخسارة المحتملة عن كل أسبوع توقف، ثم تقرر كم تنفق على المرونة: مورد بديل، أو مخزون احتياطي، أو عقود شحن طويلة الأجل، أو مسارات بديلة.

PRACTICE · Practice questions

1. Your main supplier of a critical component is in one city and supplies 90% of your needs. Lost margin if supply stops is $200,000 per week; probability of a 4-week disruption in a year estimated at 5%. Qualifying a second supplier costs $60,000 a year in lost discounts and admin. Should you do it?
2. Containers repeatedly incur demurrage because B/Ls arrive late under L/C payment. Propose two solutions.

ANSWER · Answers to the practice questions

1. Expected annual loss ≈ 5% × 4 × 200,000 = **$40,000** — below $60,000 on expected value. But the impact (a $800,000 loss) may be unacceptable for the company's liquidity and customer relationships; consider partial measures (qualify a backup supplier without regular volume, hold 3–4 weeks of safety stock), or proceed if risk appetite is low.
2. Ask the supplier to present documents faster and courier originals to the bank by express; request an L/C allowing presentation of one original B/L to the buyer directly (with the rest through the bank) — riskier for the bank; use electronic B/Ls where the banks and carrier support them; or obtain a bank guarantee to the carrier for release before original B/Ls arrive (letter of indemnity backed by a bank — only with careful risk assessment).

CHAPTER 73

Simulation: Importing from China into the Middle East

LEARNING OBJECTIVES · In this chapter

A complete training simulation: supplier quotations and selection, purchase order, payment, freight, insurance, Saudi customs duties and VAT, conformity certification, port charges, inland transport, financing costs, landed cost by product with freight allocation, pricing and sensitivity. **All companies, quotations, rates and regulatory fees are fictional assumptions for training; verify current Saudi customs tariffs, VAT and product-conformity requirements before real use.**

73.1 Background

Najd Home Appliances Trading (fictional), Riyadh, Saudi Arabia, wants to import three small-appliance products for its wholesale customers (retail chains and hypermarkets). Functional currency SAR (pegged at 3.75 per USD). Purchases are quoted in USD.

SKU	Product	Quantity	Carton volume	Carton weight
A	Microwave oven 20 L	900	0.060 m³	12.0 kg
B	Electric kettle 1.7 L	2,400	0.012 m³	1.3 kg
C	Air fryer 4 L	1,000	0.045 m³	5.0 kg

Total volume = 54.0 + 28.8 + 45.0 = **127.8 m³ → two 40ft high-cube containers** (usable ~68 m³ each). Total gross weight = 10,800 + 3,120 + 5,000 = **18,920 kg**.

73.2 Step 1 — Supplier quotations (microwave example)

	Supplier X (Ningbo)	Supplier Y (Foshan)	Supplier Z (Ningbo)
Quoted price	$38.50 FOB Ningbo	$36.90 EXW factory	$41.00 CFR Dammam
Add to reach CFR Dammam (per unit)	Freight $1.97	Inland + export clearance $1.10 + freight $1.97	—
Comparable CFR Dammam	**$40.47**	**$39.97**	**$41.00**
Payment terms	30% deposit, 70% against copy B/L	50% deposit, 50% before shipment	30% deposit, 70% L/C at sight
Lead time	40 days	50 days	45 days
Certifications for Saudi market	Has prior conformity registrations	Not yet	Has
Factory audit result	Good	Acceptable	Good

Decision: Supplier X — price within 1.3% of the cheapest, better payment terms than Y, shorter lead time, existing conformity experience, and the buyer keeps control of freight under FOB. Suppliers for B and C were selected similarly; for simplicity, assume all three SKUs are bought from Supplier X (a consolidated order).

73.3 Step 2 — Purchase order and payment

Item	Value (USD)
A: 900 × $38.50	34,650.00
B: 2,400 × $6.20	14,880.00
C: 1,000 × $24.00	24,000.00
Total FOB Ningbo, Incoterms® 2020	**73,530.00**
30% deposit (day 0)	22,059.00
70% balance against copy of B/L (day 45)	51,471.00

Controls: verified supplier bank details by phone; PO states specifications, plug type and voltage (Saudi standards), Arabic/English labelling, packing and marking, conformity responsibilities, inspection right, warranty (2% free spare parts), delay penalty (0.5% per week, max 5%), and arbitration clause. Optional third-party pre-shipment inspection: $320 (included below in local costs as "inspection").

73.4 Step 3 — Freight and insurance

- Ocean freight Ningbo → Dammam (King Abdulaziz Port), all-in incl. bunker: **2 × $2,100 = $4,200** (booked by the buyer's forwarder; origin THC paid by the seller under FOB).
- Cargo insurance ICC (A) incl. war and strikes: 0.15% × 110% × (FOB + freight) = 0.0015 × 1.1 × 77,730 = **$128.25**.
- **CIF value = 73,530 + 4,200 + 128.25 = $77,858.25** (Saudi customs value basis).

73.5 Step 4 — Allocation of freight and insurance to products

Freight is allocated by **volume** (it fills containers); insurance by **value**.

	A	B	C	Total
Volume share	42.25%	22.54%	35.21%	100%
FOB value share	47.12%	20.24%	32.64%	100%
FOB value	34,650.00	14,880.00	24,000.00	73,530.00
Freight (by volume)	1,774.65	946.48	1,478.87	4,200.00
Insurance (by value)	60.44	25.96	41.86	128.26
CIF value	**36,485.09**	**15,852.44**	**25,520.73**	**77,858.26**

73.6 Step 5 — Customs duty and VAT (illustrative rates)

	A (duty 10%)	B (duty 10%)	C (duty 5%)	Total
Customs duty	3,648.51	1,585.24	1,276.04	6,509.79
Import VAT 15% on (CIF + duty) — recoverable	6,020.04	2,615.65	4,019.52	12,655.21

Duty rates are illustrative; actual rates depend on the HS classification in the current Saudi tariff schedule (GCC common external tariff with national schedules). VAT is recoverable by the VAT-registered importer but must be funded at clearance.

73.7 Step 6 — Local costs

Cost	Amount (USD)	Allocation basis
Product conformity certificates (3 × $350) and shipment certificates (3 × $150)	1,500	Per SKU ($500 each)
Delivery order and port fees ($350) + destination THC (2 × $280)	910	Volume
Customs broker ($300) + pre-shipment inspection share/customs handling ($100)	400	Volume
Trucking Dammam → Riyadh (2 × $850) + unloading (2 × $120)	1,940	Volume
Bank transfer charges (2 × $45)	90	Value
Financing cost (8% p.a. on cash tied up until customers pay, see below)	2,020	Value
Total local and financing costs	**6,860**	

Financing estimate: deposit $22,059 tied up ~135 days (until customers pay) → $653; balance + freight + insurance $55,799 for ~90 days → $1,101; duties and local costs ~$10,250 for ~63 days → $142; VAT $12,655 for ~45 days until recovered → $125. Total ≈ **$2,020**.

73.8 Step 7 — Landed cost by product

	A Microwave	B Kettle	C Air fryer	Total
CIF	36,485.09	15,852.44	25,520.73	77,858.26
Duty	3,648.51	1,585.24	1,276.04	6,509.79
Conformity certificates	500.00	500.00	500.00	1,500.00
Port/DO/THC	384.48	205.11	320.41	910.00
Broker/handling	169.00	90.16	140.84	400.00
Trucking/unloading	819.65	437.28	683.07	1,940.00
Bank charges	42.41	18.21	29.38	90.00
Financing	951.82	408.85	659.33	2,020.00
Landed cost (excl. VAT)	**43,000.96**	**19,097.29**	**29,129.80**	**91,228.05**
Units	900	2,400	1,000	
Landed cost per unit (USD)	**47.78**	**7.96**	**29.13**	
Per unit (SAR)	179.18	29.85	109.24	
Uplift vs FOB	+24.1%	+28.4%	+21.4%	+24.1%

The kettle carries the highest uplift because it is bulky relative to its value and absorbs a fixed $500 certification cost over a low-value shipment.

73.9 Step 8 — Pricing

Target wholesale gross margin 30% on landed cost: Price = landed ÷ 0.70.

	A	B	C
Wholesale price (USD)	68.26	11.37	41.61
Wholesale price (SAR, excl. VAT)	255.98	42.64	156.04

> **COMMON MISTAKES · What happens with naive pricing**
>
> If the sales team priced at FOB × 1.30 (a common shortcut), the microwave would sell at $50.05: margin on landed cost = (50.05 − 47.78) ÷ 50.05 = **4.5%**, not 30%. For the kettle, $8.06 → margin 1.2%.

> Landed-cost discipline is the difference between a profitable import business and a busy one that loses money.

73.10 Step 9 — Timeline

Day	Event	Documents
0	PO signed; deposit paid	PO, pro forma invoice, transfer confirmation
5–20	Conformity registration for products completed (with supplier test reports)	Product certificates
40	Pre-shipment inspection at factory	Inspection report
42	Containers stuffed; VGM; export clearance by seller	Packing list, export declaration
45	Loaded on board; B/L issued; balance paid against B/L copy; original B/L couriered (or telex release after payment)	Commercial invoice, B/L, certificate of origin, insurance certificate
47	Shipment certificates issued	Shipment conformity certificates
68	Vessel arrives Dammam	Arrival notice
69	Delivery order; declaration filed; duty and VAT paid	Import declaration, DO
71	Release; trucks dispatched	Release order
72	Delivery to Riyadh warehouse; receiving inspection	GRN
75–135	Sales to customers on 30-day terms; collections	Invoices

73.11 Step 10 — Sensitivity and risks

Scenario	Effect on total landed cost	Effect on blended margin at planned prices
Freight rate doubles to $4,200/container	+$4,200 freight, +$7 insurance, +~$347 duty → **+$4,554 (+5.0%)**	Gross margin falls from 30.0% to ~26.5%
7-day customs hold (demurrage $120/container/day after free time)	+$1,680	−1.3 pts
Supplier delivers 3 weeks late	Missed Ramadan season → potential discounting of 10% on 50% of stock	−~5 pts
HS classification of air fryer at 10% instead of 5%	+$1,276	−1.0 pt
2% of microwaves defective	Supplier provides spares; repair cost ~$600	−0.5 pt

Controls: freight contract with fixed rates for the season; documents checklist and pre-clearance; delay penalties and early production slots; advance HS ruling; inspection; warranty clause.

SUMMARY · Lessons from the simulation

- Compare suppliers on a landed basis, not on headline price; payment terms, lead time and compliance capability matter.
- Allocate freight by the driver that consumes it (volume for containers), insurance by value; fixed per-SKU costs hit small shipments hard.
- Duty and VAT must be funded at clearance; VAT recovery affects cash, not cost.
- Price from landed cost; shortcuts based on FOB destroy margins.

- Build the timeline around regulatory steps (conformity certification) and seasonal demand.

PRACTICE · Practice questions

1. Recompute the kettle's landed cost per unit if the order quantity doubles to 4,800 units (assume the extra volume fits in the same two containers, freight unchanged in total but reallocated by volume, and certification cost unchanged).
2. Which allocation basis would you use for: (a) trucking costs; (b) bank charges; (c) insurance; (d) customs broker fees per declaration line?
3. The supplier offers DAP Riyadh at an extra $6,800 total over FOB. Compare with the buyer's own costs for freight, insurance, port, trucking and unloading and advise.

ANSWER · Answers to the practice questions

1. New volumes: A 54, B 57.6, C 45 → 156.6 m³ (would actually require checking container capacity — 136 m³ usable — so a third container would be needed; the simplification shows the method). Using the simplification: B's volume share = 57.6 ÷ 156.6 = 36.78% → freight 1,544.83; FOB B = 29,760 (value share 33.67% of 88,410) → insurance ≈ 128.25 × (FOB+freight share) ≈ 51.6 (approx.); CIF ≈ 31,356; duty 10% ≈ 3,136; certificates 500; volume-based local costs 36.78% × (910 + 400 + 1,940) = 1,195; value-based ~33.67% × (90 + financing ~2,400) ≈ 838 → landed ≈ 37,025 → **≈ $7.71 per unit** (vs $7.96): fixed costs spread over more units. In reality, check whether a third container is needed — if so, the cost per unit could rise.
2. (a) Volume (or per container); (b) value or per transfer; (c) value; (d) number of declaration lines/SKUs, or value if line counts are similar.
3. Buyer's own costs for those elements: freight 4,200 + insurance 128 + DO/port/THC 910 + trucking/unloading 1,940 = **$7,178**. DAP at +$6,800 is cheaper by ~$378 and shifts transit risk to the seller (the seller bears risk to Riyadh). However, the buyer loses control of carrier choice and timing, must still clear customs and pay duties (on a CIF value that customs will reconstruct), and DAP pricing may hide destination charges. Check the supplier's forwarder capability and whether destination THC and DO fees are truly included before accepting.

PART XV

Insurance and Risk Management

التأمين وإدارة المخاطر

Every business decision carries risk. Insurance is one way to deal with it — by transferring certain losses to an insurer for a premium — but it is only one of four basic responses: avoid, reduce, transfer, or retain. This part explains how insurance works, the commercial lines most businesses buy, how enterprise risk management organises all risks, and how four different businesses — an importer, a logistics company, a manufacturer and a commodity trader — make concrete coverage decisions.

LEARNING OBJECTIVES · Learning objectives for Part XV

- Explain the mechanics of insurance: premiums, deductibles, limits, exclusions, underwriting, claims and reinsurance.
- Describe the main commercial insurance lines and what each does and does not cover.
- Apply enterprise risk management: identification, assessment, treatment, registers, controls and business continuity.
- Decide which risks to insure, reduce, transfer contractually or retain, for different business models.

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CHAPTER 74

Insurance Fundamentals

LEARNING OBJECTIVES · In this chapter

The principles of insurance contracts, how premiums are set, the structure of a policy, the claims process, and the roles of brokers and reinsurers.

74.1 How insurance works

Insurance pools risks: many policyholders pay premiums; the few who suffer insured losses are compensated. It works best for risks that are **numerous, independent, measurable, accidental and not catastrophic for the insurer**. Uninsurable or hard-to-insure risks include ordinary business risks (losing customers, bad strategy), deliberate acts, and risks that are certain to happen.

74.2 Core principles (common-law tradition; local laws vary)

Principle	Meaning	Practical effect
Insurable interest	The insured must suffer a financial loss if the event occurs	You cannot insure a competitor's warehouse
Utmost good faith / duty of disclosure	Material facts must be disclosed to the insurer	Non-disclosure can void cover; some jurisdictions now apply a "fair presentation" duty
Indemnity	Compensation restores the insured's position, no profit	Payment limited to actual loss (except agreed-value policies)
Contribution	Several policies covering the same loss share it	No double recovery
Subrogation	Insurer takes over the insured's rights against third parties	Don't waive rights against carriers or contractors without insurer consent
Proximate cause	The dominant cause of loss determines cover	Disputes when several causes combine

74.3 Anatomy of a policy

Element	Description
Schedule / declarations	Insured, period, sum insured, limits, premium, deductible
Insuring clause	What is covered (perils, property, liabilities)
Definitions	Meaning of key terms
Exclusions	What is not covered
Conditions	Obligations of the insured (maintenance, security, notification, cooperation)
Warranties (insurance sense)	Promises that must be strictly complied with (e.g., sprinklers maintained); breach consequences vary by law
Endorsements	Modifications to the standard wording
Limits and sub-limits	Maximum payable; specific caps for certain losses
Deductible / excess / self-insured retention	Insured's share of each loss

74.4 Pricing: premiums

Premium = expected losses + expenses + cost of capital + profit, adjusted for market conditions (insurance markets alternate between **soft** markets — ample capacity, falling prices — and **hard** markets — scarce capacity, rising prices and stricter terms).

> **WORKED EXAMPLE · Premium logic (fictional property risk)**
>
> Warehouse and stock value $10m. Insurer's data: annual probability of a fire loss 0.4%, average severity 40% of value → expected loss = 0.4% × 40% × 10m = **$16,000**. Adding expenses (30%), capital and profit loading → premium ≈ $25,000–30,000 (rate ≈ 0.25–0.30%). Sprinklers and good housekeeping may halve the probability, reducing the premium; a deductible of $50,000 removes frequent small losses and lowers it further.

74.5 Underwriting

Underwriters evaluate: nature of the business, loss history (usually 5 years), risk surveys (fire protection, security), financial strength, contracts (liabilities assumed), geography (natural catastrophe exposure), management quality. Better information and risk management lead to better terms.

74.6 Claims

1. **Notify** promptly per policy conditions.
2. **Mitigate** further loss (the insured must act as if uninsured).
3. **Preserve evidence**; do not admit liability to third parties without insurer consent.
4. **Loss adjuster** investigates cause and quantum.
5. **Document**: invoices, inventory records, accounts, photos, surveys, police reports.
6. **Settlement**: indemnity minus deductible, subject to limits, average (underinsurance) and exclusions.
7. **Subrogation**: insurer pursues responsible parties.

> **COMMON MISTAKES · Underinsurance and the average clause**
>
> If a property is insured for less than its value, many policies reduce claims proportionally. Stock worth $2m insured for $1.5m; loss $400,000 → payment = 400,000 × 1.5/2.0 = $300,000 (before deductible). Keep sums insured updated, especially in inflation or with seasonal stock peaks (use declaration or seasonal-increase clauses).

74.7 Intermediaries and reinsurance

- **Brokers** represent the insured, design programmes, place risks with insurers, and assist with claims; paid by commission or fees.
- **Agents** represent insurers.
- **Reinsurers** insure insurers (treaty or facultative), enabling local insurers to write large risks.
- **Captive insurers** are insurance companies owned by a corporate group to insure its own risks — used by large groups to retain predictable risks efficiently and access reinsurance.
- **Takaful** is a Sharia-compliant cooperative alternative: participants contribute to a fund managed by an operator under wakala or mudaraba models.

> بالعربية — مبادئ التأمين

يقوم التأمين على تجميع المخاطر: يدفع كثيرون أقساطاً صغيرة لتعويض القلة الذين يتعرضون لخسائر مؤمن عليها. ومن مبادئه: المصلحة التأمينية، ومنتهى حسن النية والإفصاح عن الوقائع الجوهرية، والتعويض بقدر الخسارة دون ربح، والحلول (حق شركة التأمين في الرجوع على المتسبب). وتتكون الوثيقة من نطاق التغطية والاستثناءات والشروط والتحمّل وحدود التعويض. ويجب الانتباه إلى قاعدة النسبية عند نقص مبلغ التأمين عن القيمة الحقيقية.

PRACTICE · Practice questions

1. Stock valued $5m insured for $4m; deductible $25,000; fire loss $800,000. Estimate the claim payment if average applies.
2. Why might a company choose a higher deductible?
3. A company signs a contract with a logistics provider waiving all claims against it. What insurance problem may arise?

ANSWER · Answers to the practice questions

1. Proportion 4/5 = 80% → 800,000 × 0.8 = 640,000 − 25,000 = **$615,000** (order of applying deductible and average depends on wording).
2. Lower premium; small losses are predictable and cheaper to self-fund; aligns incentives for loss prevention; may access better terms in a hard market.
3. The insurer loses subrogation rights; if the policy prohibits such waivers or requires consent, the company may prejudice its cover. Check policies before signing waiver clauses, or obtain a waiver-of-subrogation endorsement.

CHAPTER 75

Commercial Insurance Lines

LEARNING OBJECTIVES · In this chapter

The main commercial insurance products, what they cover, key exclusions, and who needs them.

Line	Covers	Key exclusions / issues	Typical buyers
Commercial property (all risks or named perils)	Damage to buildings, machinery, stock from fire, flood, storm, theft, etc.	Wear and tear, gradual deterioration, war; natural catastrophe sub-limits	Everyone with physical assets
Business interruption (BI)	Lost gross profit and increased cost of working after insured property damage	Usually requires a covered physical loss; indemnity period must be long enough; supplier/customer extensions needed for dependency	Manufacturers, retailers, warehouses
Engineering / machinery breakdown	Sudden breakdown of machinery; construction all risks (CAR) and erection all risks (EAR)	Wear and tear, maintenance failures	Manufacturers, contractors
General / public liability	Third-party bodily injury and property damage	Contractual liabilities beyond law, professional errors, pollution (often)	All businesses
Product liability	Injury or damage caused by products	Product recall costs (separate cover), faulty workmanship itself	Manufacturers, importers, distributors
Professional indemnity (E&O)	Financial loss to clients from professional negligence	Fraud; contractual guarantees	Consultants, brokers, IT firms, forwarders (as intermediaries)
Directors and officers (D&O)	Personal liability of directors/officers for wrongful acts; defence costs	Fraud, intentional illegal acts; prior known claims	Companies of all sizes, especially with investors or lenders
Employers' liability / workers' compensation	Employee injuries (often mandatory)	Varies by law	Employers
Motor fleet	Vehicles: third-party (often mandatory) and own damage	Unlicensed drivers, misuse	Logistics, sales fleets
Marine cargo	Goods in transit by sea, air, road; Institute Cargo Clauses (A) all risks, (B), (C) named perils; war and strikes clauses	Insufficient packing, inherent vice, delay, wilful misconduct, sanctions	Importers, exporters, traders
Marine hull and machinery	Physical damage to vessels	Wear and tear, unseaworthiness with privity of owner	Shipowners
Protection & indemnity (P&I)	Shipowners' third-party liabilities (crew, pollution, cargo, wreck removal)	Provided mainly by mutual P&I clubs	Shipowners, charterers (charterers' liability)
Freight forwarders' / carriers' liability	Liability as forwarder or carrier under trading conditions	Contracted liabilities above standard terms	Forwarders, NVOCCs, truckers
Aviation	Hull, liability, cargo for aircraft operators; airport and aviation products liability	War (separate), wear and tear	Airlines, operators
Trade credit insurance	Non-payment by buyers (commercial and political)	Disputed debts, sales beyond limits, late reporting	Exporters, distributors, traders

Line	Covers	Key exclusions / issues	Typical buyers
Political risk insurance	Expropriation, currency inconvertibility and transfer restriction, political violence, contract frustration	Devaluation (not covered — only inconvertibility), pre-existing issues	Investors, contractors, traders in emerging markets
Cyber	First-party costs (incident response, data restoration, BI, ransom where lawful) and third-party liability	War exclusions, poor security warranties, unpatched systems	All digital businesses
Crime / fidelity	Employee theft, fraud, social engineering (often sub-limited)	Losses discovered after policy, poor controls	Companies with payment flows
Environmental impairment	Pollution liabilities and clean-up	Known conditions	Industrial sites
Key person / life	Loss of key individuals	—	Owner-managed businesses, startups
Warranty and indemnity (W&I)	Breaches of warranties in M&A	Known issues, forward-looking statements	Buyers/sellers in acquisitions

PROFESSIONAL INSIGHT · Business interruption — the cover that saves companies

After a fire, the property policy rebuilds the warehouse; the BI policy pays the profits lost and the extra costs (temporary premises, overtime) during the **indemnity period**. Choose the indemnity period realistically (12 months is often too short for rebuilding and recovering customers — 18–24 months is common for manufacturers) and calculate gross profit as the insurer's wording defines it (often revenue minus variable costs, not accounting gross profit).

PRACTICE · Practice questions

1. Which policies respond: (a) a customer is injured by a faulty kettle your company imported; (b) your server is encrypted by ransomware; (c) your buyer in another country becomes insolvent owing $300,000; (d) your factory is flooded and closed for 5 months?
2. A forwarder's standard trading conditions limit liability to 2 SDR per kg. A customer asks the forwarder to accept full value liability. What insurance does the forwarder need?

ANSWER · Answers to the practice questions

1. (a) Product liability; (b) cyber insurance (first-party costs and BI) and possibly crime cover for any fraud; (c) trade credit insurance; (d) property (damage) and business interruption (lost gross profit, extra expense).
2. Forwarder's liability insurance extended to the higher contractual liability (or the customer should buy cargo insurance — usually more efficient); the forwarder should charge for the extra liability or decline.

CHAPTER 76

Enterprise Risk Management, Risk Registers, Controls and Business Continuity

LEARNING OBJECTIVES · In this chapter

A structured approach to all risks: frameworks, risk identification and assessment, treatment options, risk appetite, registers and heat maps, key risk indicators, internal controls, and business continuity planning.

76.1 Frameworks

ISO 31000:2018 (risk management guidelines) and **COSO Enterprise Risk Management — Integrating with Strategy and Performance (2017)** are the most widely used frameworks. Both emphasise integrating risk into strategy and decisions rather than maintaining a separate compliance exercise.

76.2 The risk management process

1. **Establish context:** objectives, stakeholders, risk appetite.
2. **Identify risks:** workshops, interviews, process walkthroughs, historical losses, industry studies, PESTLE, supplier and customer mapping.
3. **Analyse:** causes, consequences, likelihood, impact, existing controls.
4. **Evaluate:** compare with appetite; prioritise.
5. **Treat:** choose responses and actions.
6. **Monitor and review:** KRIs, incidents, audits.
7. **Communicate and report:** to management, board, risk committees.

76.3 Risk treatment options

Response	Description	Examples
Avoid	Stop the activity	Exit a sanctioned market; refuse a high-risk customer
Reduce (mitigate)	Lower likelihood or impact	Sprinklers, dual sourcing, controls, training, hedging
Transfer	Shift financial consequences to others	Insurance, contracts (indemnities, Incoterms, liability caps), L/Cs, credit insurance, outsourcing
Retain (accept)	Bear the risk consciously	Deductibles, self-insurance, minor risks
Exploit (for upside risks)	Take advantage of opportunities	Expanding where competitors retreat

76.4 Assessment: likelihood × impact

Likelihood	Description	Indicative probability (per year)
1 Rare	Exceptional	< 2%
2 Unlikely	Could occur	2–10%
3 Possible	Might occur	10–30%
4 Likely	Will probably occur	30–70%

Likelihood	Description	Indicative probability (per year)
5 Almost certain	Expected	> 70%

Impact	Financial (example for a $50m revenue company)	Other dimensions
1 Minor	< $50k	Minor customer complaints
2 Moderate	$50–250k	Short disruption
3 Major	$250k–1m	Loss of a key customer, regulatory warning
4 Severe	$1–5m	Prolonged disruption, fines
5 Critical	> $5m	Threat to survival, licence loss

Score = likelihood × impact (1–25); heat map colours guide prioritisation. Note the weakness: high-impact, low-likelihood risks can be under-prioritised by multiplication — boards should review all risks with critical impact regardless of score.

76.5 A risk register extract (fictional importer-distributor)

TRAINING SIMULATION · FICTIONAL DOCUMENT · NOT A VALID INSTRUMENT

Risk register (training extract)

ID	Risk	Cause	Consequence	L	I	Score	Existing controls	Actions	Owner	KRI
R1	Key supplier failure	Single-source supplier in one country	Stock-outs; lost sales $0.4m/month	2	4	8	6 weeks safety stock	Qualify second supplier by Q3	Head of Procurement	Supplier OTIF; supplier credit score
R2	FX loss on EUR purchases	30% of purchases in EUR, prices fixed in local currency	Margin erosion	4	3	12	Ad hoc forwards	Hedging policy: 75% of firm orders	CFO	Unhedged exposure %
R3	Customer default	Concentration: top 3 = 40% of receivables	Bad debt up to $1.2m	2	4	8	Credit limits	Credit insurance; monthly review	Credit Manager	Overdue > 60 days %
R4	Payment fraud (BEC)	Email-based bank detail changes	Loss up to $250k per incident	3	3	9	Dual approval	Call-back verification; training; crime insurance	Treasury Manager	Bank-detail change requests
R5	Warehouse fire	Electrical faults; stock density	Stock loss $6m; BI	1	5	5	Sprinklers; property/BI insurance	Thermal imaging inspections; review sums insured	Operations Director	Inspection findings open
R6	Customs non-compliance	Misclassification	Back-duties, penalties, delays	3	3	9	Broker review	Advance rulings; annual audit	Supply Chain Manager	Customs queries per month

Training simulation.

76.6 Internal controls and the three lines

Controls are the day-to-day mechanisms that reduce risk (see Chapter 26 (Volume 2)). The **three lines model** assigns roles: management owns risks and controls; risk and compliance functions set frameworks and challenge; internal audit provides independent assurance to the board.

76.7 Key risk indicators (KRIs)

KRIs are metrics that signal rising risk early: supplier OTIF trend, overdue receivables %, unhedged FX exposure, staff turnover in critical roles, number of system outages, customs holds per month, inventory aged > 180 days. Each KRI has thresholds (green/amber/red) and escalation rules.

76.8 Business continuity management

Element	Content
Business impact analysis (BIA)	Critical processes, maximum tolerable downtime, recovery time objectives (RTO), recovery point objectives (RPO) for data
Continuity strategies	Alternate sites, backup suppliers, remote work, manual workarounds, IT disaster recovery
Plans	Crisis management team, communication plan, recovery procedures
Testing	Tabletop exercises, failover tests; updates after changes
Standards	ISO 22301 (business continuity management systems)

بالعربية — إدارة المخاطر المؤسسية

تهدف إدارة المخاطر المؤسسية إلى التعامل المنظم مع جميع المخاطر: تحديدها وتحليلها وتقييمها ومعالجتها ومتابعتها. وخيارات المعالجة أربعة رئيسية: التجنب، والتخفيف، والنقل (بالتأمين أو العقود)، والقبول. ويُوثَّق ذلك في سجل المخاطر مع تحديد المسؤول عن كل خطر ومؤشرات المخاطر الرئيسية التي تنذر مبكراً بارتفاعه. أما خطة استمرارية الأعمال فتحدد كيف تستمر العمليات الحرجة عند وقوع أزمة كبيرة.

PRACTICE · Practice questions

1. Score and propose treatments for: “Main IT system (ERP) unavailable for more than 3 days due to ransomware” for a distributor processing 500 orders per day.
2. Give two KRIs for credit risk and two for supply-chain risk.
3. Why can a risk heat map mislead a board?

ANSWER · Answers to the practice questions

1. Likelihood 3 (possible, given ransomware prevalence), impact 4 (lost sales, delayed deliveries, recovery cost, reputation) → score 12. Treatments: reduce (MFA, patching, EDR, offline/immutable backups, segmentation, staff training), prepare (incident response plan, tested restore with RTO < 24 h, manual order process), transfer (cyber insurance with BI), and monitor KRIs (patching backlog, phishing test failure rate).
2. Credit: % of receivables overdue > 60 days; concentration of top-5 customers; credit-limit excesses. Supply chain: supplier OTIF; days of safety stock for critical items; % of spend single-sourced.
3. Scores compress complex risks into one number; multiplication underweights rare catastrophic risks; subjective ratings vary; correlations between risks (one event triggering several) are ignored; it can create a false sense of precision.

CHAPTER 77

Insurance and Risk Decisions for Four Businesses

LEARNING OBJECTIVES · In this chapter

Four fictional case studies showing how risk profiles drive insurance programmes and other risk treatments: an importer-distributor, a logistics company, a manufacturer and an international commodity trader.

77.1 Case 1 — Importer-distributor of consumer electronics

Profile: Revenue $60m; imports from Asia (FOB terms); one central warehouse (stock up to $12m before peak season); 400 retail customers on 45-day terms; 120 employees.

Risk	Decision	Rationale
Goods in transit (buyer's risk from loading under FOB)	**Marine cargo open cover**, ICC (A) + war/strikes, all shipments declared monthly; deductible $5,000	High value, theft-prone goods; frequent shipments
Warehouse fire/flood/theft	**Property all risks** with stock declaration clause (peak values), **BI** 18-month indemnity	Single site concentration
Customer insolvency	**Trade credit insurance** on top 80 customers; smaller accounts self-insured with tight limits	Receivables ~$7m; concentration in chains
Product liability (imported electricals)	**Product liability** $5m limit; supplier contracts with indemnities and supplier insurance certificates	Importer often treated as producer under consumer-protection laws
FX (USD purchases, local sales)	Not insurance — **treasury hedging policy**	Market risk
Payment fraud	**Crime policy** with social-engineering extension; call-back controls	BEC is frequent
Customs penalties	**Retain**; reduce through compliance controls	Generally uninsurable fines

77.2 Case 2 — Logistics company (trucking, warehousing, forwarding)

Profile: 80 trucks, two 3PL warehouses holding customers' goods, freight forwarding as NVOCC.

Risk	Decision
Liability for customers' goods in its care	**Carrier's and warehouse keeper's liability** insurance aligned with trading conditions (e.g., CMR limits on road; contractual limits in warehousing) — avoid accepting unlimited liability in contracts
NVOCC liability under house B/Ls	**Freight forwarders' liability** / NVOCC E&O, including errors in documents and customs
Trucks	**Motor fleet** third-party (mandatory) and own damage on newer vehicles; older trucks self-insured for own damage
Warehouses	Property (own buildings/equipment) + **customers' goods** covered by the customers' own property insurance as agreed in contracts
Employees	Workers' compensation/employer's liability per law; medical as required
Cyber	**Cyber** cover: tracking systems and customs interfaces are critical
Customs bonds	**Customs guarantees** from banks (not insurance) for bonded operations

Key lesson: the logistics company's exposure is mostly **liability for others' goods** — controlled by contract terms (limits, exclusions, time bars) and matched by liability insurance.

77.3 Case 3 — Manufacturer (plastics packaging)

Profile: One plant with 12 extrusion lines; revenue $45m; customers in food industry with strict quality requirements.

Risk	Decision
Fire/explosion at plant	Property all risks (reinstatement value), **BI 24 months** (lead time for replacement machines ~10 months + customer recovery)
Machinery breakdown	**Machinery breakdown** cover + BI extension for breakdown; critical spare parts stock
Product contamination/defect affecting customers' food	**Product liability** and **product recall** insurance; quality certifications (e.g., food-safety standards)
Supplier (resin) disruption	Not insurable cost-effectively → dual sourcing, 4 weeks' stock, **contingent BI** for named key supplier if available
Environmental (spills)	Environmental impairment liability (often excluded from general liability)
Directors' liability (bank-financed, private equity minority)	**D&O**

77.4 Case 4 — International commodity trader (grains)

Profile: Trades 1.2m tonnes per year of wheat and corn between the Black Sea, South America and North Africa/Middle East; uses L/Cs and open account; finances via transactional trade-finance lines.

Risk	Decision
Cargo loss/damage at sea	**Marine cargo** (ICC A or grain-specific trade clauses), including war risk for certain routes; often required by financing banks (assigned as loss payee)
Quality/quantity disputes	Contract terms (final at loading per independent inspection), inspection companies, not insurance
Charterer's liabilities (damage to vessel, pollution, cargo claims)	**Charterer's liability** insurance (P&I-type cover for charterers)
Buyer default on open account sales	**Trade credit insurance** and/or confirmed L/Cs; bank requires credit-insured receivables for financing
Political risk (export bans, contract frustration, transfer restrictions)	**Political risk insurance** for exposures in high-risk countries
Price risk	**Not insurance** — hedging with futures/swaps (Part XI)
Fraud (fake warehouse receipts, double financing)	Collateral managers with their own liability insurance; KYC; **crime** cover has limits — mainly controls
Sanctions	Compliance controls; insurance will not pay for sanctioned voyages

> **PROFESSIONAL INSIGHT · The general pattern**
>
> Insurance is efficient for **accidental, measurable and independent** losses (fire, cargo damage, liability claims, buyer insolvency). Market risks (price, FX, interest rates) are managed with **hedging**; strategic and operational risks with **management and controls**; contractual risks with **contract terms**. A professional builds a combined programme and makes sure the pieces fit — for example, that Incoterms, cargo insurance, carrier liability and sales-contract claims procedures align.

PRACTICE · Practice questions

1. For the importer in Case 1, estimate a suitable BI indemnity period and justify it.
2. The commodity trader is asked by its bank to assign cargo insurance and credit insurance policies. Why?
3. Name three risks that insurance generally cannot cover and the alternative treatment for each.

ANSWER · Answers to the practice questions

1. 12–18 months: finding and fitting out a new warehouse, restocking (2–3 months' lead time from Asia), and regaining customers who switched suppliers during disruption; peak-season dependence argues for 18 months.
2. The goods and receivables financed are the bank's collateral; assignment (or loss-payee clauses) ensures insurance proceeds go to the bank if cargo is lost or a buyer defaults, protecting repayment.
3. Price/FX movements → hedging; loss of market share due to competition → strategy and product development; fines for deliberate regulatory breaches → compliance controls; poor business decisions → governance and analysis.

SUMMARY · Part XV summary

- Insurance transfers accidental, measurable losses for a premium; read limits, deductibles, exclusions and conditions.
- Business interruption, product liability, cargo, credit and cyber covers are often the most important for trading and manufacturing companies.
- ERM organises all risks — avoid, reduce, transfer, retain — with registers, KRIs and clear ownership.
- Different business models need different programmes; align insurance with contracts, Incoterms and financing requirements.

Index to Volume 5

Bold page numbers mark the main term-card definition.